# Methodological Extensions to SNB Monetary Policy Transmission Analysis

Companion Document to the Master Thesis

*Monetary Policy Transmission in Switzerland:*
*Exchange Rate and Inflation Responses to Rate Cuts, 2000–2025*

February 2026

This document presents seven methodological extensions to the main thesis analysis.

All results are generated by the `extensions/` pipeline and are self-contained.

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# 1 Introduction

This companion document presents seven methodological extensions to the main thesis, which analyzes Swiss National Bank (SNB) monetary policy transmission using a purging regression approach combined with local projections (Jordà, 2005). The main analysis identifies monetary policy shocks as residuals from a purging regression of policy rate changes on macroeconomic controls, then traces their dynamic effects on the EUR/CHF exchange rate and CPI inflation over 12-month horizons.

These extensions serve three purposes. First, they strengthen the identification of monetary policy shocks by employing high-frequency intraday data as an external instrument (Section 2) and by decomposing shocks into pure monetary and information components (Section 6). Second, they enhance statistical inference through wild bootstrap confidence intervals (Section 4) and placebo/falsification tests (Section 5). Third, they investigate nonlinearities and heterogeneity in transmission through smooth transition local projections (Section 3), additional outcome variables (Section 7), and asymmetry tests by shock sign and size (Section 8).

All extensions use the same data and shock identification from the main pipeline and are implemented as self-contained R scripts in the `extensions/R/` directory.

# 2 High-Frequency Identification

## 2.1 Motivation

The purging regression approach identifies monetary policy shocks by controlling for the central bank's systematic response to macroeconomic conditions. A complementary identification strategy uses high-frequency asset price movements in narrow windows around policy announcements as external instruments (Gertler and Karadi, 2015; Stock and Watson, 2018). The key insight is that in a sufficiently tight window, asset price changes primarily reflect the policy surprise rather than other macroeconomic news.

## 2.2 Methodology

We use intraday EUR/CHF exchange rate changes in a $[-1\text{h}, +1\text{h}]$ window around the 09:30 CET SNB announcement time as an instrument for the monthly monetary policy shock in a two-stage least squares (2SLS) local projection framework. The first stage regresses the purging-regression residual on the high-frequency surprise and controls:

$$\text{mp_shock}_t = \alpha + \gamma \cdot \text{hf_surprise}_t + \boldsymbol{X}_t'\boldsymbol{\delta} + u_t \tag{1}$$

The second stage at each horizon $h$ is:

$$\Delta y_{t+h} = \beta_h \cdot \widehat{\text{mp_shock}}_t + \boldsymbol{X}_t'\boldsymbol{\phi}_h + \varepsilon_{t+h} \tag{2}$$

## 2.3 Results

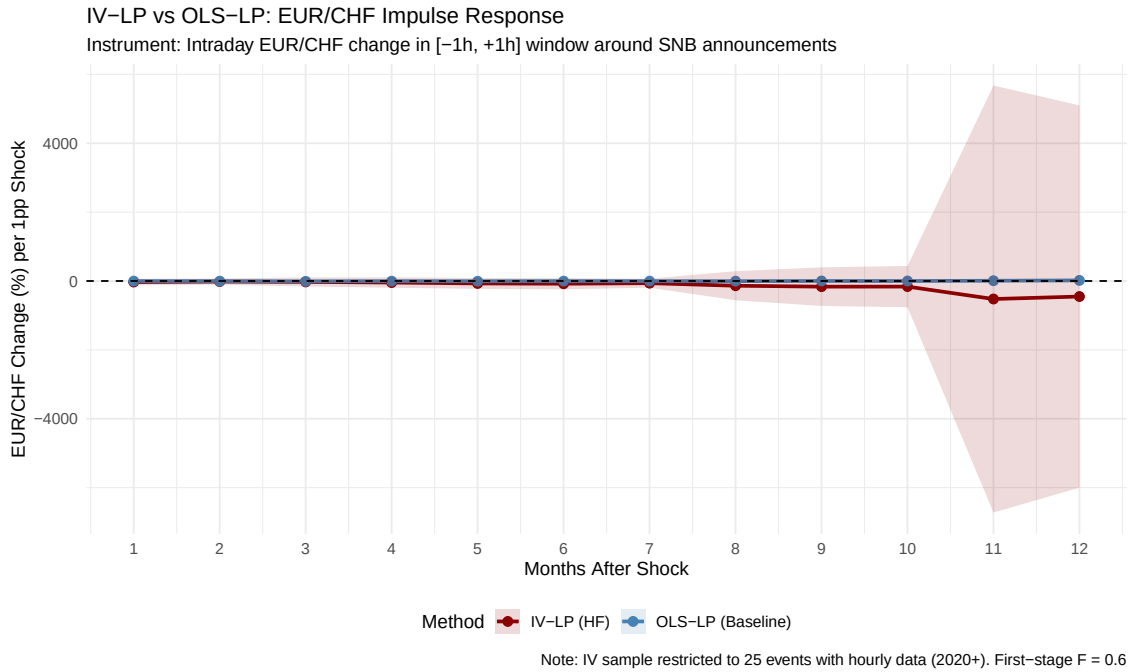



Figure 1: IV-LP vs OLS-LP: EUR/CHF impulse response comparison. The IV estimates use intraday EUR/CHF changes as an instrument for the monetary policy shock.

Table 1: First-Stage Regression: HF EUR/CHF Surprise as Instrument for MP Shock

| Variable | Estimate | SE | t.stat | p.value |
|---|---|---|---|---|
| (Intercept) | -0.0970 | (0.0731) | -1.33 | 0.200 |
| hf_surprise | 0.1279 | (0.1623) | 0.79 | 0.440 |
| yield_2y | 0.0604 | (0.0294) | 2.06 | 0.054 |
| vix | 0.0004 | (0.0009) | 0.44 | 0.668 |
| kof_lag1 | 0.0012 | (0.0007) | 1.80 | 0.088 |
| cpi_yoy_lag1 | -0.0229 | (0.0116) | -1.97 | 0.064 |

Table 2: IV-LP vs OLS-LP: EUR/CHF Response Comparison

| Method | h = 1 | h = 3 | h = 6 | h = 12 |
|---|---|---|---|---|
| IV-LP (HF) | -30.61 (47.50) | -26.86 (66.92) | -79.30 (82.95) | -451.07 (2831.50) |
| OLS-LP (Baseline) | -2.79 (2.12) | -9.68*** (1.89) | -3.62 (3.25) | 17.48*** (5.02) |

## 2.4 Discussion

The high-frequency identification provides an important cross-check on the baseline purging regression approach. The sample is restricted to events from 2020 onward where hourly EUR/CHF data is available, yielding a small number of observations. The first-stage F-statistic provides a diagnostic for instrument strength. If IV and OLS estimates are qualitatively similar, this supports the validity of the baseline identification; systematic differences may indicate endogeneity bias in the OLS estimates or weak instrument problems in the IV approach. Given the small sample, these results should be interpreted as suggestive rather than definitive.

# 3 Smooth Transition Local Projections

## 3.1 Motivation

The main analysis uses a hard regime dummy (positive vs. negative interest rates) to capture potential nonlinearities in monetary policy transmission. However, the transition between regimes is unlikely to be discrete. Following the smooth transition regression framework of Auerbach and Gorodnichenko (2013) and Tenreyro and Thwaites (2016), we replace the binary indicator with a logistic transition function that allows for gradual regime shifts.

## 3.2 Methodology

Define the transition function:

$$G(z_t; \gamma, c) = \frac{1}{1 + \exp\left(\gamma \cdot (z_t - c)\right)} \tag{3}$$

where $z_t$ is the policy rate, $\gamma$ controls the transition speed (higher $\gamma$ approximates the hard threshold), and $c = 0\%$ is the threshold. The smooth transition LP is:

$$\Delta y_{t+h} = \beta_h^{\text{low}} \cdot \text{mp_shock}_t \cdot G(z_t) + \beta_h^{\text{high}} \cdot \text{mp_shock}_t \cdot (1 - G(z_t)) + \boldsymbol{X}_t' \boldsymbol{\phi}_h + \varepsilon_{t+h} \tag{4}$$

We estimate this for $\gamma \in \{1, 3, 5, 10\}$ to assess sensitivity to the transition speed.

## 3.3 Results

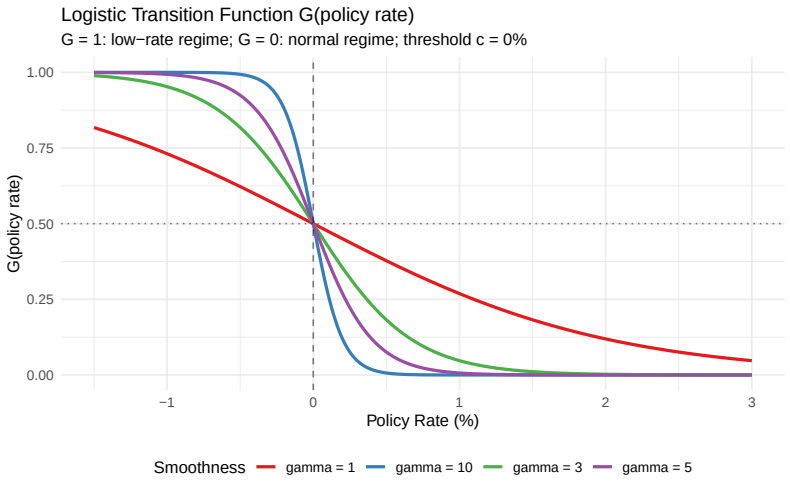



Figure 2: Logistic transition function $G$(policy rate) for different values of the smoothness parameter $\gamma$. Higher $\gamma$ approximates the hard regime dummy.

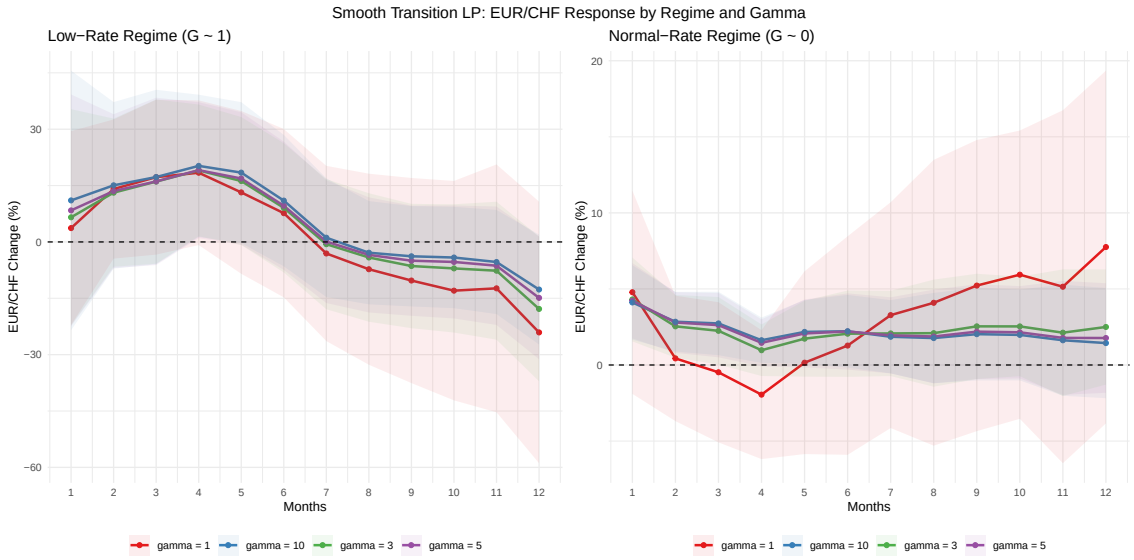



Figure 3: Smooth transition LP: EUR/CHF impulse responses in the low-rate regime ($G \approx 1$) and normal-rate regime ($G \approx 0$) for different $\gamma$ values.

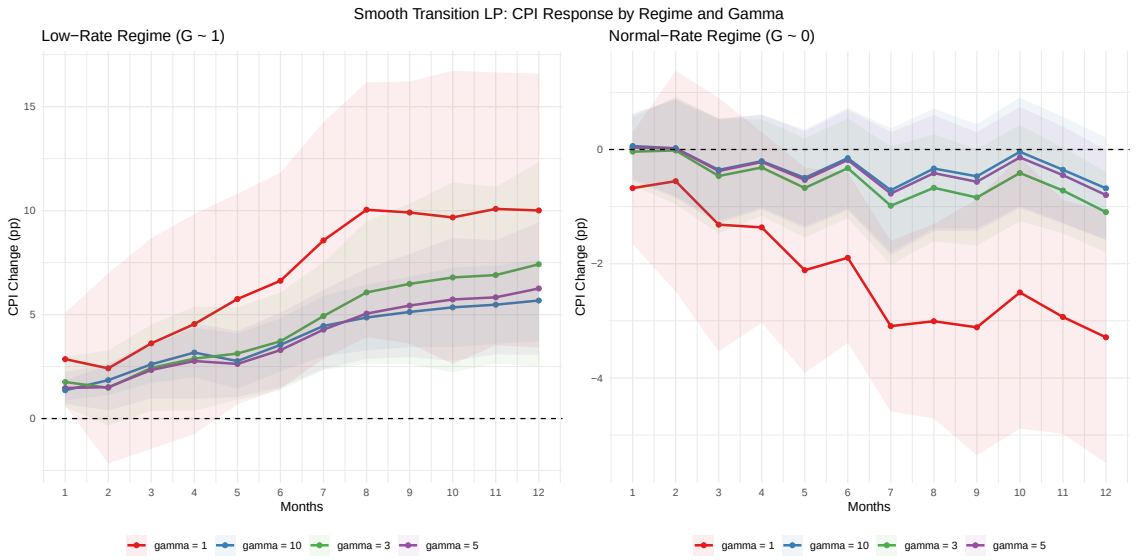



Figure 4: Smooth transition LP: CPI impulse responses by regime and transition speed.

Table 3: Smooth Transition LP: EUR/CHF Response (gamma = 5, c = 0%)

| Horizon | Low-Rate Regime | Normal-Rate Regime | Difference |
|---|---|---|---|
| 1 | 8.39 (15.75) | 4.22 (1.26) | 4.17 |
| 3 | 16.11 (11.37) | 2.62 (1.08) | 13.49 |
| 6 | 9.63 (8.71) | 2.20 (1.28) | 7.43 |
| 12 | -14.89 (8.33) | 1.77 (1.84) | -16.66 |

## 3.4 Discussion

The smooth transition framework reveals whether the regime dependence found in the main analysis is robust to the specification of the regime boundary. If results are stable across $\gamma$ values, this suggests the findings are not driven by the arbitrary binary split. Conversely, sensitivity to $\gamma$ would indicate that the transition between regimes matters for the estimated transmission mechanism. As $\gamma \to \infty$, the smooth transition model converges to the hard-threshold specification.

# 4 Wild Bootstrap Confidence Intervals

## 4.1 Motivation

Local projection standard errors are typically computed using the Newey and West (1987) heteroskedasticity and autocorrelation consistent (HAC) estimator. However, HAC inference can perform poorly in small samples, which is relevant for our analysis given the limited number of policy rate changes. The Rademacher wild bootstrap provides an alternative that has better finite-sample properties (Gonçalves and Kilian, 2004).

## 4.2 Methodology

For each horizon $h$:

1. Estimate the baseline LP and compute the observed $t$-statistic for the shock coefficient.

2. Under $H_0 : \beta_h = 0$, estimate the restricted model and obtain fitted values $\hat{y}_t^r$ and residuals $\hat{e}_t^r$.

3. For $b = 1, \ldots, B$ (with $B = 999$):

   - Draw Rademacher weights $w_t \in \{-1, +1\}$ with equal probability.
   - Construct bootstrap outcome: $y_t^* = \hat{y}_t^r + \hat{e}_t^r \cdot w_t$.
   - Estimate the LP on bootstrap data and compute the bootstrap $t$-statistic $t_b^*$.

4. The bootstrap $p$-value is $\hat{p} = B^{-1} \sum_{b=1}^{B} \mathbf{1}(|t_b^*| \geq |t_{\text{obs}}|)$.

5. Percentile-$t$ bootstrap CIs: $[\hat{\beta} - q_{0.975}^* \cdot \hat{\sigma},\ \hat{\beta} - q_{0.025}^* \cdot \hat{\sigma}]$.

## 4.3 Results

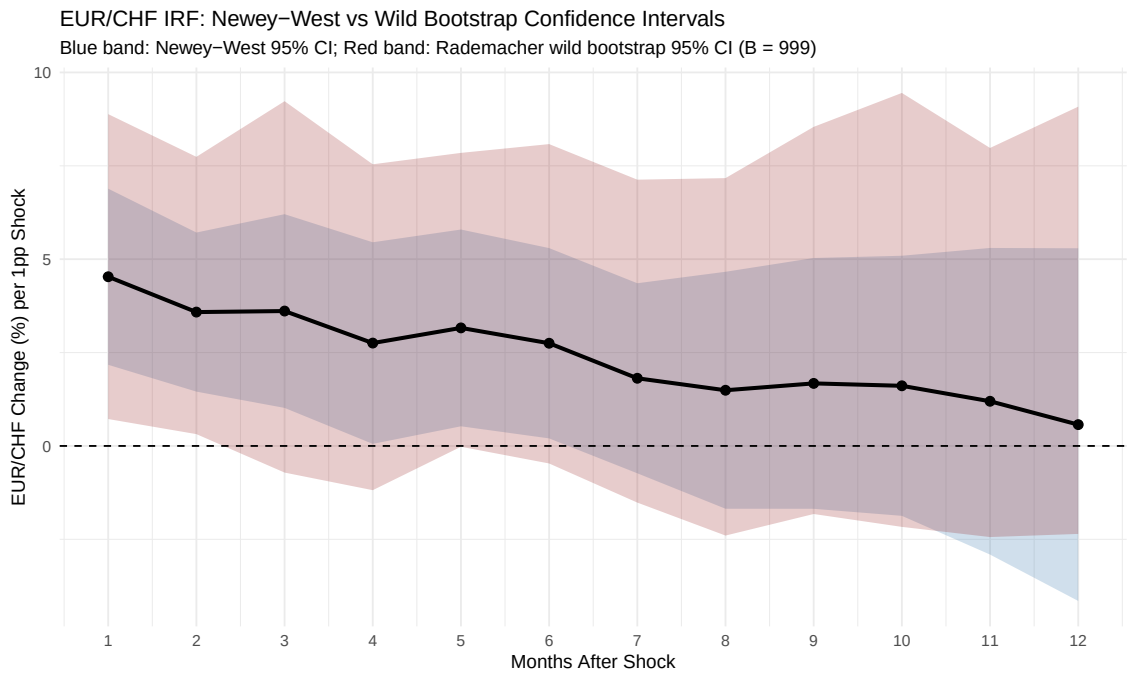



Figure 5: EUR/CHF IRF with Newey-West (blue) and wild bootstrap (red) 95% confidence intervals.

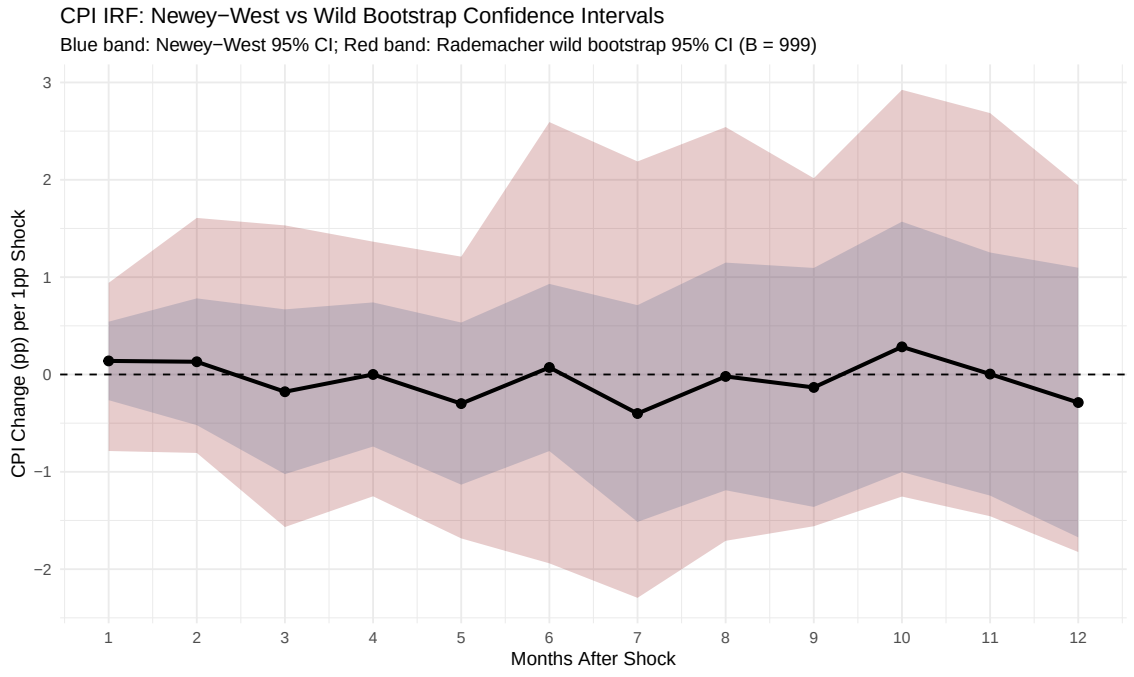



Figure 6: CPI IRF with Newey-West and wild bootstrap confidence intervals.

Table 4: Wild Bootstrap vs Newey-West: EUR/CHF Response (B = 999)

| Horizon | Coeff. | NW 95% CI | NW p-val | Bootstrap 95% CI | Boot. p-val |
|---|---|---|---|---|---|
| 1 | 4.53 | [2.17, 6.89] | 0.000 | [0.72, 8.88] | 0.025 |
| 3 | 3.61 | [1.02, 6.20] | 0.006 | [-0.71, 9.23] | 0.128 |
| 6 | 2.75 | [0.20, 5.30] | 0.034 | [-0.47, 8.08] | 0.293 |
| 12 | 0.57 | [-4.15, 5.29] | 0.813 | [-2.36, 9.08] | 0.848 |

### 4.4 Discussion

Comparing the two sets of confidence intervals reveals whether asymptotic Newey-West inference provides adequate coverage in our sample. If bootstrap CIs are substantially wider, the asymptotic approximation may be overly optimistic, and some findings that appear significant under Newey-West may not survive bootstrap inference. Conversely, similar CIs would increase confidence in the baseline results.

## 5 Placebo and Falsification Tests

### 5.1 Motivation

A well-identified monetary policy shock should only produce significant effects when actual policy changes occur. If the same LP specification yields significant results when applied to "no-change" meetings where the SNB held rates constant, this would cast doubt on the identification strategy (Ramey, 2016).

### 5.2 Methodology

We construct two types of placebo tests:

1. **Fitted-value placebo**: On meeting dates where $\Delta r_t = 0$, we use the predicted component from the purging regression (i.e., the fitted value $\hat{\Delta} r_t$, representing the anticipated policy action) as a "shock." If identification is valid, anticipated actions should not produce significant exchange rate or inflation responses.

2. **Permutation test**: We randomly shuffle the shock assignment across time periods ($N = 500$ permutations) and re-estimate the LPs. The permutation $p$-value is the fraction of randomized coefficients exceeding the actual estimate in absolute value.

## 5.3 Results

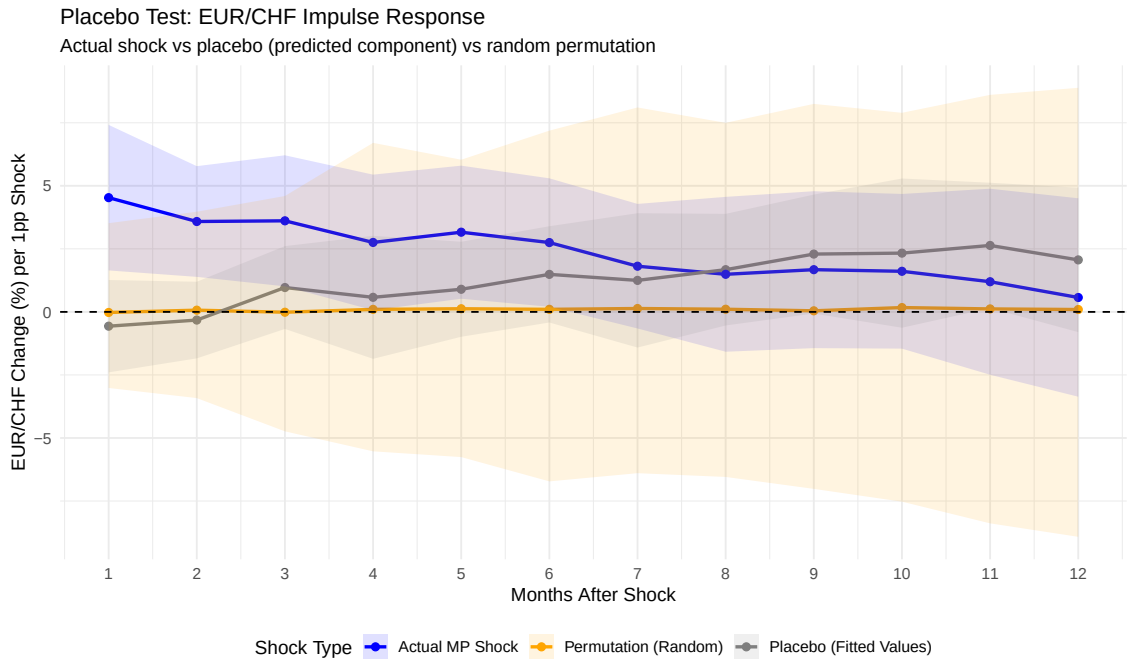



Figure 7: Placebo test for EUR/CHF: actual shock vs. placebo (fitted values) vs. random permutation.

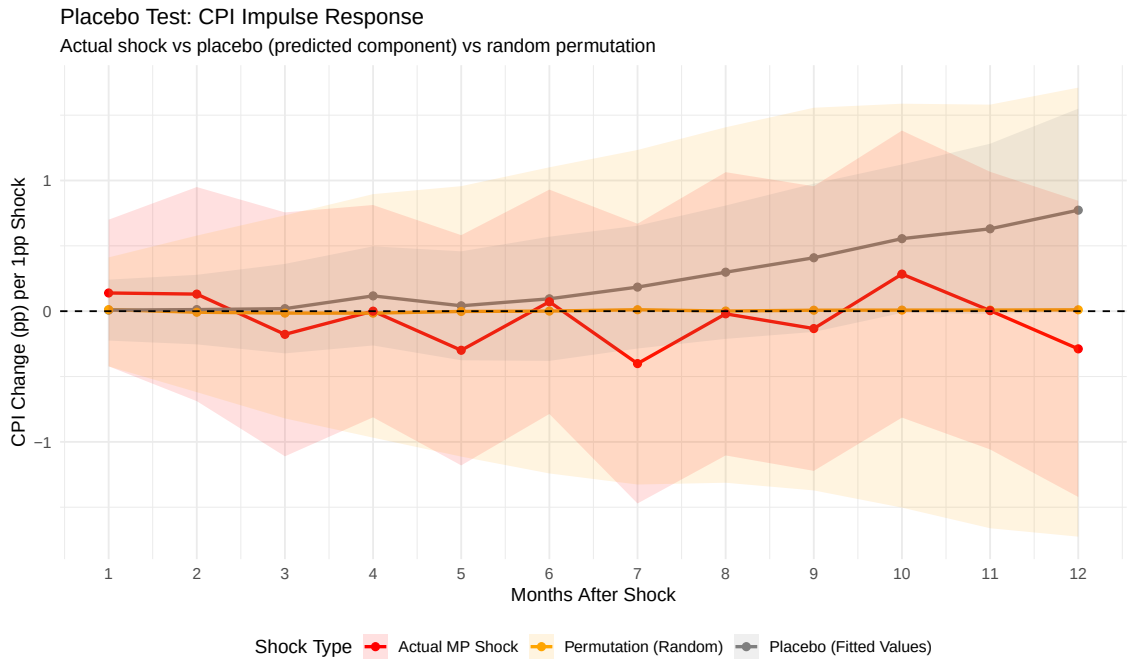



Figure 8: Placebo test for CPI: actual shock vs. placebo vs. random permutation.

Table 5: Placebo and Permutation Tests: Comparison of Responses

| Variable | h = 1 | h = 3 | h = 6 | h = 12 |
|---|---|---|---|---|
| Actual (FX) | 4.53 | 3.61 | 2.75 | 0.57 |
| Placebo (FX) | -0.57 | 0.97 | 1.49 | 2.06 |
| Perm. p (FX) | 0.018 | 0.116 | 0.344 | 0.878 |
| Actual (CPI) | 0.14 | -0.18 | 0.07 | -0.29 |
| Placebo (CPI) | 0.01 | 0.02 | 0.09 | 0.77 |
| Perm. p (CPI) | 0.544 | 0.654 | 0.908 | 0.750 |

## 5.4 Discussion

The placebo tests provide a falsification check for the shock identification. Insignificant placebo responses (close to zero) and low permutation $p$-values for the actual shock strengthen the causal interpretation of the baseline results. If placebo shocks produce responses comparable to actual shocks, this would suggest the identification

does not adequately separate anticipated from unanticipated policy actions.

# 6 Information Effects

## 6.1 Motivation

Central bank announcements convey not only policy actions but also information about the central bank's assessment of economic conditions. Nakamura and Steinsson (2018) and Jarociński and Karadi (2020) show that disentangling "pure monetary policy" from "information" components is crucial for correctly identifying transmission mechanisms. An unexpected rate cut paired with a yield decline and CHF depreciation signals standard monetary easing, whereas a rate cut accompanied by a yield increase and CHF appreciation suggests the market learned negative economic news (information effect).

## 6.2 Methodology

We classify each nonzero MP shock month based on the co-movement of the shock and the one-month change in the 2-year government bond yield:

- **Pure MP shock**: $\text{mp_shock}_t$ and $\Delta\text{yield_2y}_t$ have the same sign (easing lowers yields, tightening raises yields).
- **Information shock**: $\text{mp_shock}_t$ and $\Delta\text{yield_2y}_t$ have opposite signs (market revises expectations about the economy).

We then create separate shock variables and estimate:

$$\Delta y_{t+h} = \beta_h^{\text{MP}} \cdot \text{shock}_t^{\text{pure}} + \beta_h^{\text{info}} \cdot \text{shock}_t^{\text{info}} + \boldsymbol{X}_t' \boldsymbol{\phi}_h + \varepsilon_{t+h} \tag{5}$$

## 6.3 Results

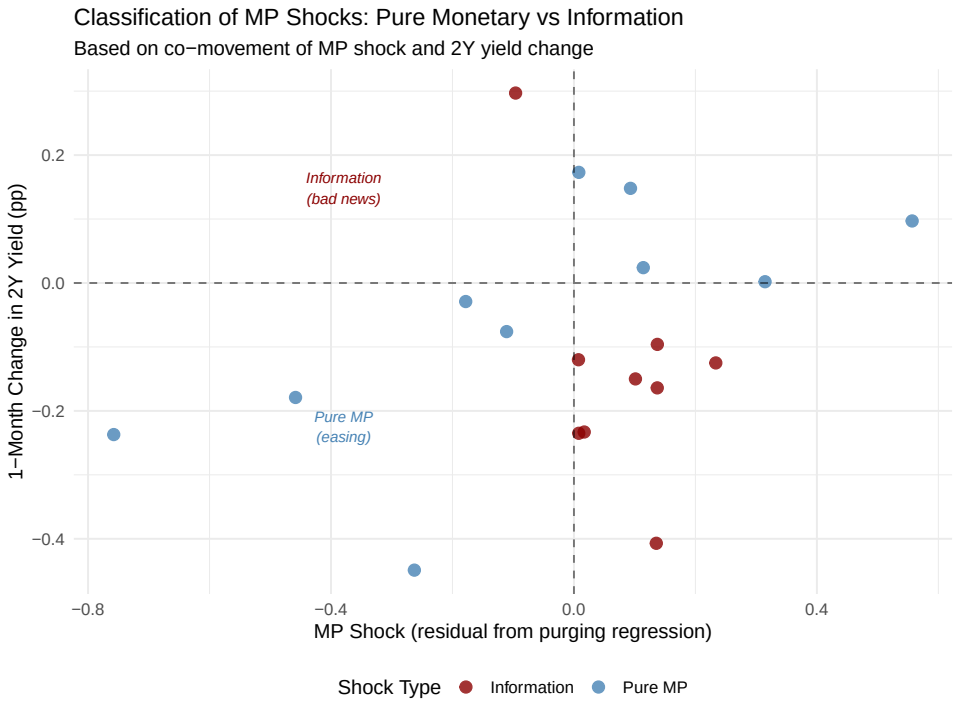



Figure 9: Classification of MP shocks into pure monetary and information components based on yield co-movement.

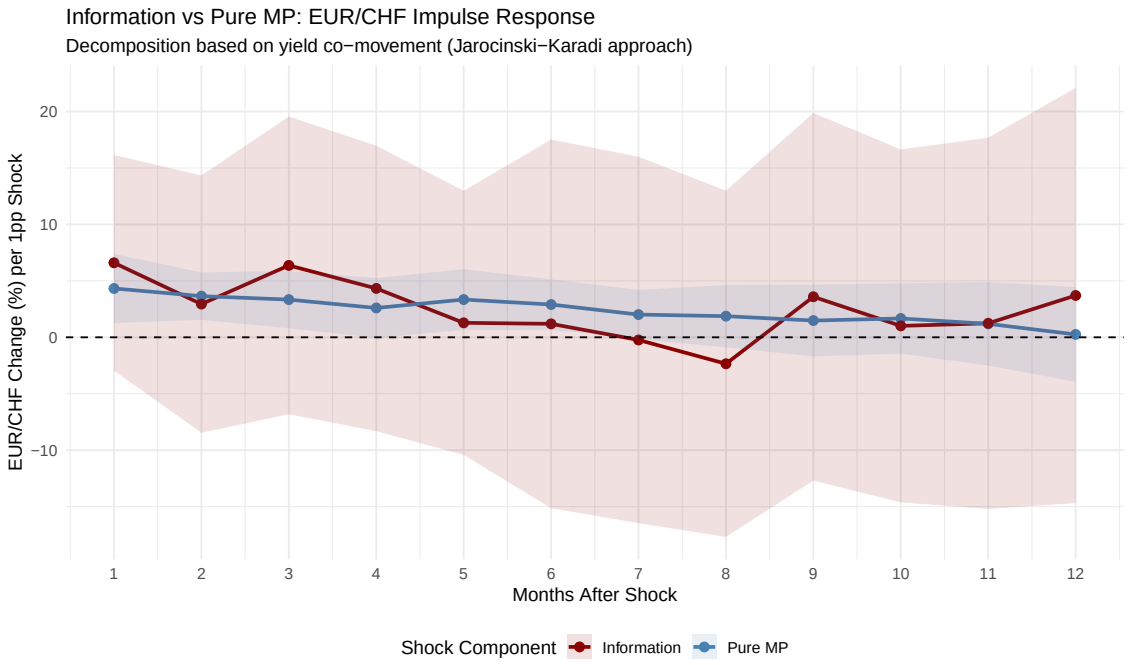



Figure 10: EUR/CHF impulse responses: pure MP shocks vs. information shocks.

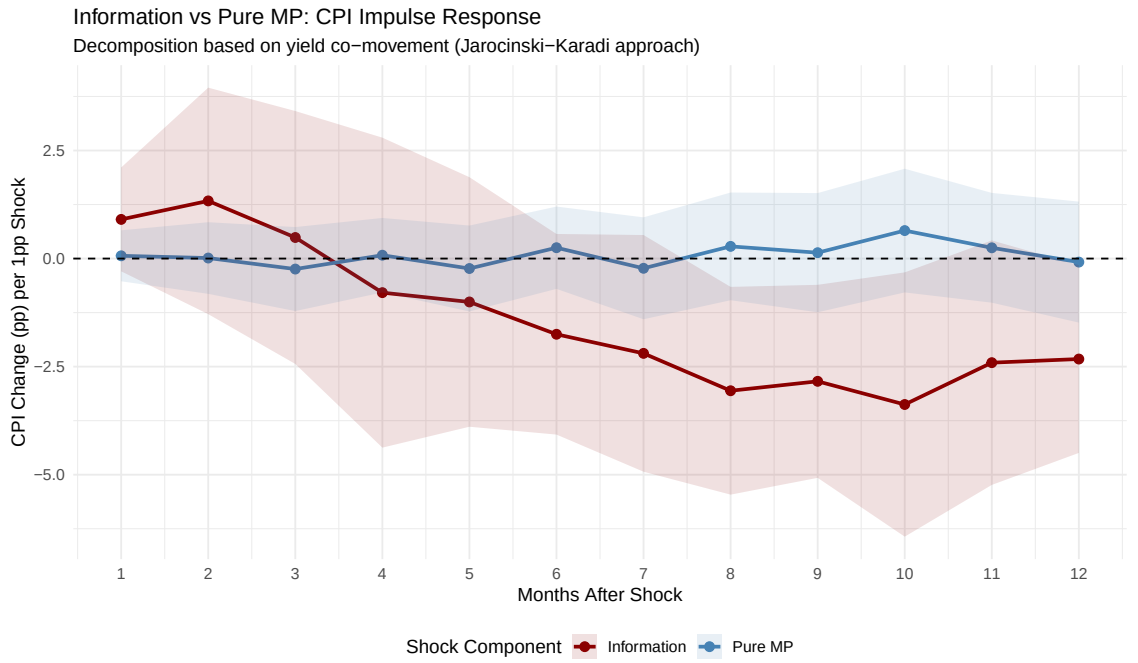



Figure 11: CPI impulse responses: pure MP shocks vs. information shocks.

Table 6: Information vs Pure MP Shocks: IRF Comparison

| Horizon | Component | EUR/CHF | CPI |
|---|---|---|---|
| 1.00 | Pure MP | 4.33 (1.56) | 0.06 (0.30) |
| 3.00 | Pure MP | 3.34 (1.30) | -0.24 (0.50) |
| 6.00 | Pure MP | 2.90 (1.14) | 0.25 (0.49) |
| 12.00 | Pure MP | 0.25 (2.14) | -0.08 (0.71) |
| 1.00 | Information | 6.60 (4.87) | 0.91 (0.61) |
| 3.00 | Information | 6.37 (6.73) | 0.49 (1.49) |
| 6.00 | Information | 1.19 (8.33) | -1.75 (1.18) |
| 12.00 | Information | 3.70 (9.38) | -2.32 (1.11) |

## 6.4 Discussion

This decomposition is a simplified version of Jarociński and Karadi (2020), adapted to the available Swiss data. The classification relies on monthly yield changes rather

than intraday movements, which introduces more noise. Nevertheless, systematic differences between pure MP and information shock responses would suggest that information effects are an important channel in SNB communication and that the baseline estimates represent a mixture of both components.

# 7 Additional Outcome Variables

## 7.1 Motivation

The main analysis focuses on EUR/CHF exchange rate and CPI inflation responses. To provide a more complete picture of the monetary transmission mechanism, we extend the LP framework to additional variables representing the yield curve channel (2-year and 10-year government bond yields) and real economic expectations (KOF Economic Barometer).

## 7.2 Methodology

For each additional outcome $y_t \in \{\text{yield_2y}, \text{yield_10y}, \text{kof}\}$, we estimate:

$$y_{t+h} - y_t = \beta_h \cdot \text{mp_shock}_t + \boldsymbol{X}_t' \boldsymbol{\phi}_h + \varepsilon_{t+h}, \quad h = 1, \ldots, 12 \tag{6}$$

using Newey-West standard errors with lag truncation $h$.

## 7.3 Results

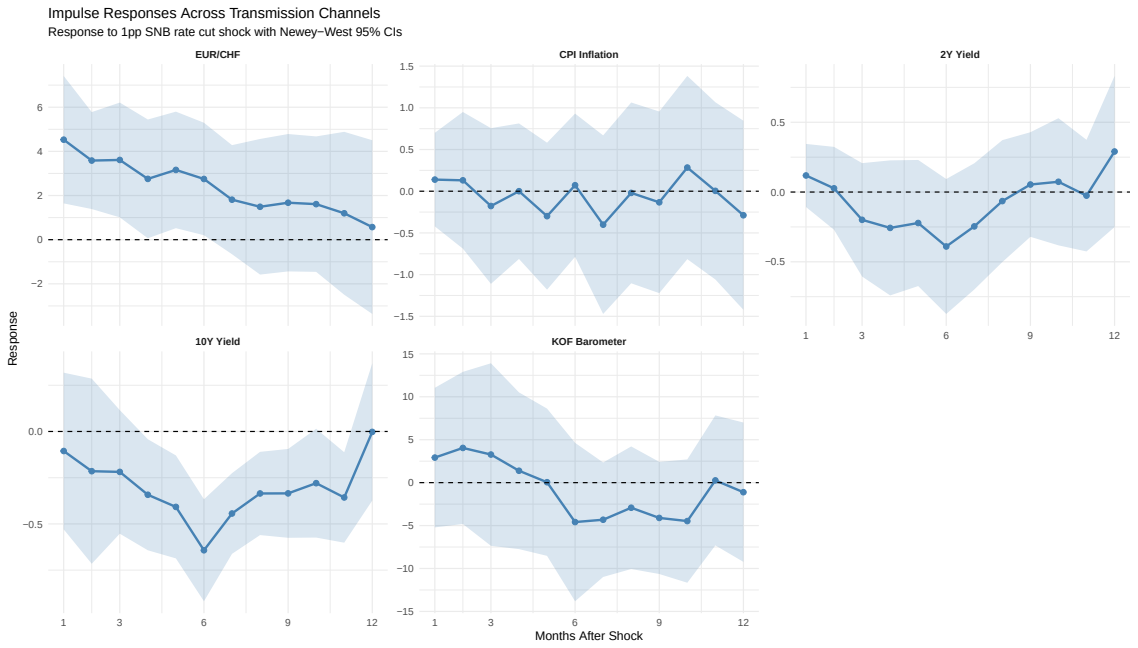



Figure 12: Multi-panel impulse responses across all transmission channels: EUR/CHF, CPI, 2Y yield, 10Y yield, and KOF barometer.

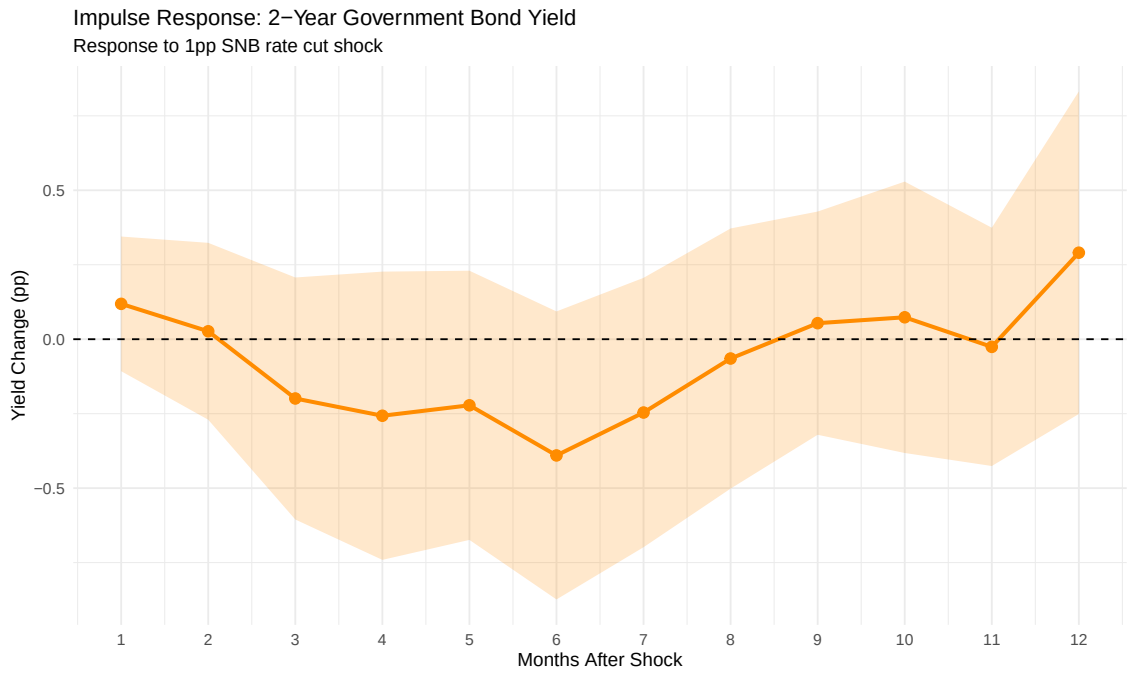



Figure 13: Impulse response: 2-year government bond yield.

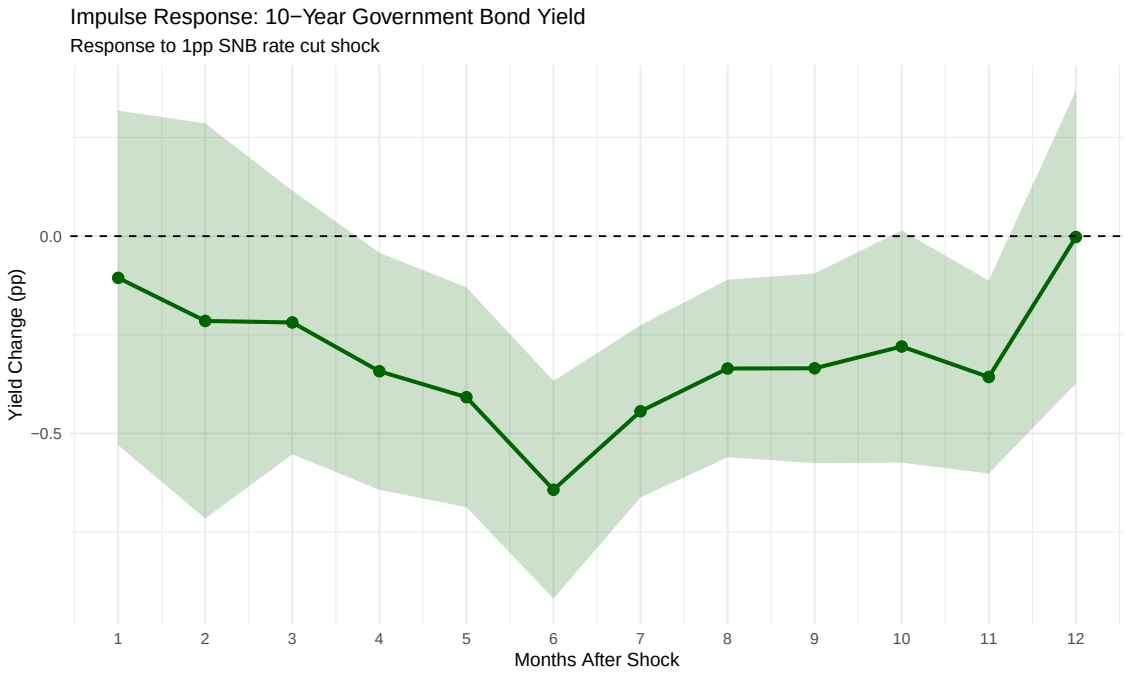



Figure 14: Impulse response: 10-year government bond yield.

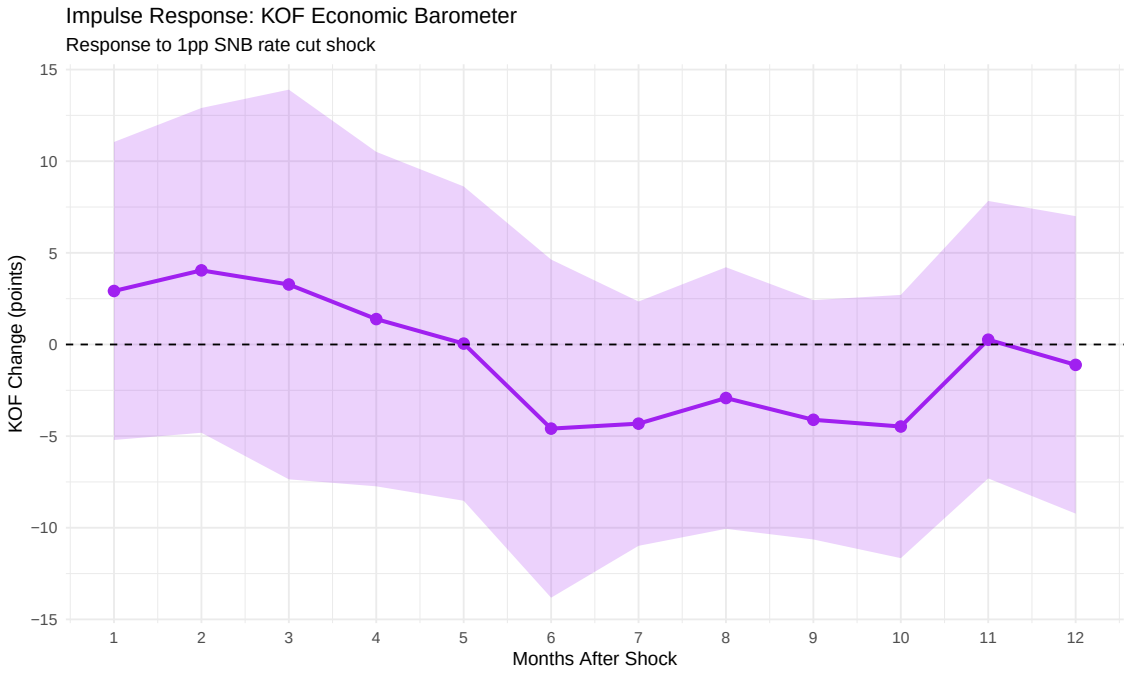



Figure 15: Impulse response: KOF Economic Barometer.

Table 7: Additional Outcome Variables: LP Impulse Responses to 1pp Rate Cut

| Horizon | 2Y Yield | 10Y Yield | KOF |
|---|---|---|---|
| 1.00 | 0.119 (0.115) | -0.106 (0.216) | 2.919 (4.148) |
| 3.00 | -0.199 (0.207) | -0.219 (0.170) | 3.273 (5.424) |
| 6.00 | -0.390 (0.247) | -0.643*** (0.141) | -4.591 (4.707) |
| 12.00 | 0.290 (0.276) | -0.002 (0.189) | -1.113 (4.141) |

### 7.4 Discussion

The yield responses trace the term structure channel of monetary policy transmission. A rate cut shock should lower short-term yields more than long-term yields, consistent with expectations channel propagation. The KOF response captures the real economy channel—whether monetary easing improves business expectations. Together with the exchange rate and inflation responses from the main analysis, these results map out the full transmission mechanism of SNB policy.

## 8 Size and Sign Asymmetry Tests

### 8.1 Motivation

Standard linear LP specifications impose symmetric responses to positive and negative shocks of equal magnitude. However, economic theory suggests several channels through which asymmetries may arise: downward nominal rigidities, different credibility of rate cuts vs. hikes, or nonlinear portfolio adjustment effects (Tenreyro and Thwaites, 2016). We test for both sign asymmetry (easing vs. tightening) and size asymmetry (large vs. small shocks).

### 8.2 Methodology

**Sign asymmetry.** We decompose the shock into easing and tightening components:

$$\Delta y_{t+h} = \beta_h^{\text{cut}} \cdot \text{shock}_t^- + \beta_h^{\text{hike}} \cdot \text{shock}_t^+ + \boldsymbol{X}_t' \boldsymbol{\phi}_h + \varepsilon_{t+h} \tag{7}$$

where $\text{shock}_t^- = \text{mp_shock}_t \cdot \mathbf{1}(\text{mp_shock}_t < 0)$ and $\text{shock}_t^+ = \text{mp_shock}_t \cdot \mathbf{1}(\text{mp_shock}_t > 0)$. A Wald test for $H_0 : \beta_h^{\text{cut}} = \beta_h^{\text{hike}}$ assesses symmetry.

**Size asymmetry.** We split shocks at the median absolute value of nonzero shocks:

$$\Delta y_{t+h} = \beta_h^{\text{large}} \cdot \text{shock}_t^L + \beta_h^{\text{small}} \cdot \text{shock}_t^S + \boldsymbol{X}_t' \boldsymbol{\phi}_h + \varepsilon_{t+h} \tag{8}$$

**Quantile regression.** As a robustness check, we estimate quantile regressions at $\tau \in \{0.25, 0.50, 0.75\}$ to characterize the conditional distribution of responses (Koenker and Bassett, 1978).

## 8.3 Results

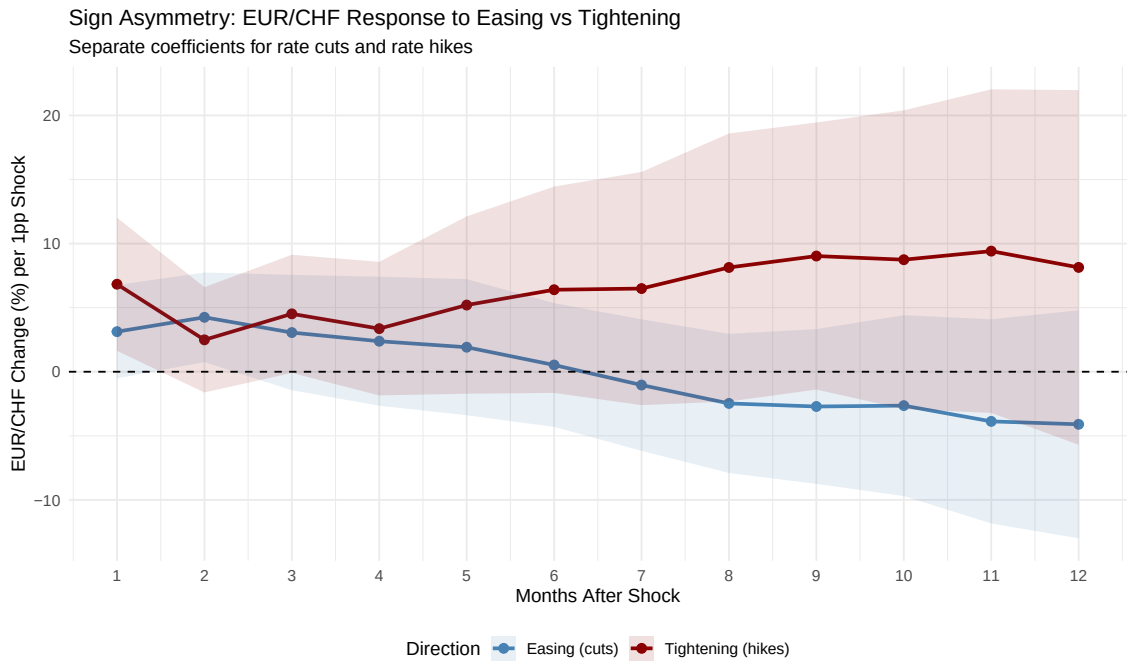



Figure 16: Sign asymmetry: EUR/CHF response to rate cuts (easing) vs. rate hikes (tightening).

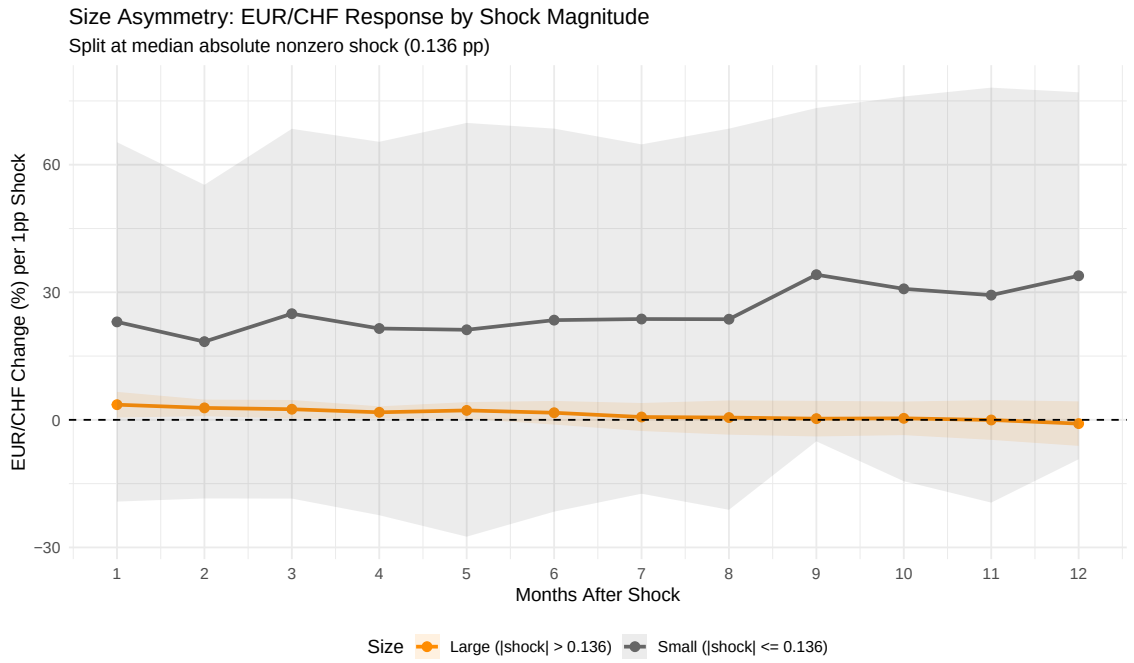



Figure 17: Size asymmetry: EUR/CHF response to large vs. small shocks.

Table 8: Asymmetry Tests: EUR/CHF Response by Shock Sign and Size

| h | Cuts | Hikes | Sign Wald | Large | Small | Size Wald |
|---|---|---|---|---|---|---|
| 1 | 3.13* (1.86) | 6.83** (2.65) | 1.08 [0.300] | 3.56** (1.54) | 23.04 (21.56) | 0.79 [0.375] |
| 3 | 3.06 (2.30) | 4.52* (2.35) | 0.14 [0.711] | 2.50** (1.11) | 24.96 (22.18) | 1.01 [0.316] |
| 6 | 0.53 (2.46) | 6.40 (4.11) | 0.94 [0.334] | 1.67 (1.42) | 23.45 (22.98) | 0.86 [0.354] |
| 12 | -4.10 (4.54) | 8.14 (7.06) | 1.25 [0.265] | -0.88 (2.67) | 33.88 (22.01) | 2.31 [0.130] |

Table 9: Quantile Regression: EUR/CHF Response at Different Quantiles

| horizon | tau = 0.25 | tau = 0.5 | tau = 0.75 |
|---|---|---|---|
| 1.00 | 2.04 (5.90) | 3.04*** (0.95) | 3.14 (2.21) |
| 3.00 | 2.29 (4.53) | 1.24 (1.89) | 0.44 (5.77) |
| 6.00 | -3.81 (3.37) | 0.49 (2.00) | 3.64** (1.56) |
| 12.00 | -6.59 (4.58) | -1.26 (3.98) | 3.63 (5.91) |

## 8.4 Discussion

Sign asymmetry is particularly relevant for the Swiss case: given the CHF's safe-haven status (Ranaldo and Söderlind, 2010), rate cuts during stress episodes may have different exchange rate effects than rate hikes during normalization periods. Size asymmetry tests whether markets respond proportionally to policy surprises or whether large shocks trigger disproportionate reactions (e.g., through portfolio rebalancing or threshold effects in market microstructure). The quantile regression results complement the mean-response analysis by revealing whether the effects are homogeneous across the conditional distribution of exchange rate changes.

# 9 Conclusion

The seven extensions presented in this document strengthen and extend the main thesis analysis along three dimensions.

**Identification.** The high-frequency IV approach (Section 2) provides an independent check on the purging regression identification, while the information effect decomposition (Section 6) disentangles pure monetary from information components in SNB announcements.

**Inference.** The wild bootstrap (Section 4) addresses potential small-sample distortions in Newey-West standard errors, and the placebo tests (Section 5) provide direct falsification evidence for the shock identification strategy.

**Heterogeneity.** The smooth transition framework (Section 3) relaxes the hard regime threshold, additional outcome variables (Section 7) map the full transmission mechanism across yields and real activity, and the asymmetry tests (Section 8) reveal whether transmission depends on the direction and magnitude of policy surprises.

Together, these extensions provide a comprehensive robustness and sensitivity analysis that complements the core findings of the main thesis.

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